

MEILAME TAYEBJEE

Applied AI Engineer ◊ Foundation Models & Production ML

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EDUCATION

ENS Paris-Saclay, MSc MVA – Mathematics, Vision, Learning 2023 – 2024

Relevant Coursework: Computational Statistics, Deep Learning for Medical Imaging, Optimization

Ensae Paris, Master of Engineering 2023 – 2024

Relevant Coursework: Bayesian Statistics, Advanced Econometrics, Machine Learning Theory

Ecole Polytechnique, Master of Engineering (X2020) 2020 – 2024

Applied Math Scientific Specialization Program

HEC Paris, Master in Management 2019 – 2024

Preparatory classes at Lycée Henri-IV (2017–2019) – Major: Math, Minor: Geopolitics & Economics

EXPERIENCE

Applied AI Engineer Sept. 2024 – current

Dual appointment, 50% research / 50% applied ML (civil service track)

Paris, France

– *AI/ML Innovation Lab, Insee (French national statistical institute)*

- Shipped *torchTextClassifiers*, an open-source PyTorch package modernizing a legacy fastText pipeline into a production MLOps system serving 10,000+ daily API requests at ~200ms inference latency.
- Developed an agentic RAG workflow for production model monitoring.
- Built a satellite imagery segmentation model for census monitoring.
- Coordinated AI/ML adoption across European National Statistical Institutes (ESSNet AIML4OS).
- Supervised efforts on explainable, trustworthy ML and on LLM-generated synthetic data for text classification.

– *Crest (Center for Research in Economics and Statistics), Inria (SODA lab)*

- Designed and trained one of the first decoder-only foundation models (300M parameters) on the complete French National Health Database (70M patients, 8B events), including distributed training infrastructure and custom medical tokenization.
- Deployed the resulting patient embeddings in a population-scale, doubly robust off-policy evaluation of France's primary-care gatekeeping policy, quantifying effects on hospitalizations, emergency visits, and mortality for public decision-makers.
- Advised by Gaël Varoquaux (Inria), Guillaume Lécué & Philippe Choné (Crest).

AI Research Intern April – August 2024

Inria (HeKA team) – Advisor: Stéphanie Allasonnière

Paris, France

- Generative model for longitudinal data via variational inference & latent diffusion models.

Graduate Research Intern April – August 2023

Massachusetts Institute of Technology (MIT, Operations Research Center)

Boston, MA, USA

Advisor: Alexandre Jacquillat

- Microtransit system modelization and optimization using linear programming.

M&A Intern June – August 2022

Rothschild & Co, Global Advisory

Paris, France

TECHNICAL SKILLS

- **ML / DL:** PyTorch, Lightning, Hugging Face Transformers, scikit-learn, PySpark
- **MLOps & LLM:** MLflow, Kubernetes, Docker, Git, Neo4j, Qdrant, LangChain, OpenAI Agents SDK

EXTRA-CURRICULAR ACTIVITIES

- **Engagement:** National debate competition winner, HEC eloquence finalist; mentored underprivileged teenagers (*Fleur de Bitume*); organizer of Polytechnique's Welcome Week (€500k budget)
- **Sports:** Tennis (competition), Soccer, Hiking
- **Languages:** French & Hindi (native), English & Spanish (advanced), Arabic (beginner)

PUBLICATIONS & WORKING PAPERS

- M. Tayebjee, L. Resende, G. Lecué, P. Choné. *Evaluating primary care gatekeeping policy via foundation models*. In preparation (2026)
- M. Tayebjee & L. Resende. *MarBLE: Markovian history tokens for compact context summarization in decoder-only Transformers*. In preparation (2026).
- M. Tayebjee & M. Lesur. *Tokenizing Longitudinal Sequences: NLP-inspired Deep Learning for Dynamic Panel Data Analysis*. Submitted to the IAOS Young Statistician Prize (2026). Extends the student-pathway component of M. Lesur's supervised Master's thesis.

TALKS

- *torchTextClassifiers: a text classification framework in PyTorch*
PyData Paris 2025 [\[Recording\]](#) & PyData Global 2025 [\[Recording\]](#) (Dec 2025)
- *Policy Evaluation using Transformer-based Medical Pathway Embeddings*
Inria SODA Seminar (Mar 2026) & CREST-INSEE Seminar (June 2026)
- *From Panel Data to Token Sequences: NLP-inspired Machine Learning for Longitudinal Categorical Data*
International Association for Official Statistics Conference (May 2026, Vilnius, Lithuania) [\[Slides\]](#)
- *From Accuracy to Trust: Practical Methods for Uncertainty-Aware Automatic Coding with ML and xAI*
International Association for Official Statistics Conference (May 2026, Vilnius, Lithuania) [\[Slides\]](#)

SUPERVISION

- Matéo Morin – Master's thesis – Feb.-July 2026
Code2Text: Synthetic textual data generation for automatic coding [\[Thesis \(in French\)\]](#)
- Julien Bernardi-Pellegrino – Master's thesis – April-July 2026
Towards Interpretable Text Classification [\[Thesis\]](#)
- Marine Lesur – Master's thesis – Jan.-Aug. 2026
A Unified Autoregressive Framework for Analyzing Pathway Determinants: Applications to the French Higher Education System. Publication in preparation (2026).